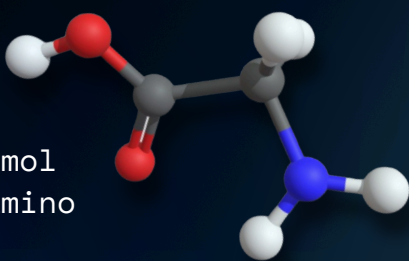




COMPOUND PROFILE

- IUPAC NAME: 2-aminoacetic acid
- MOLECULAR FORMULA: $C_2H_5NO_2$
- MOLAR MASS: 75.07 g/mol
- FUNCTIONAL GROUPS: amino + carboxylic acid



FUN FACTS

- Glycine can generate electricity from mechanical stress
- Biocompatible, biodegradable, and environmentally friendly
- Radiation can destroy glycine (as it is surprisingly vulnerable)

USES

- Protein synthesis
- Pharmaceutical applications
- Additive/flavoring
- Biochemical research
- Scientific/industrial applications

ENGINEERING APPLICATIONS

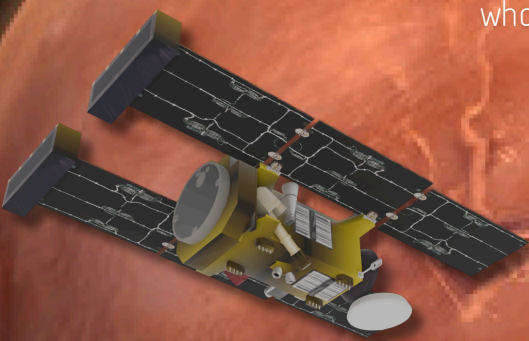
- **Crystallization & Polymorph Control:** Chemical engineers can control crystallization conditions to obtain specific glycine polymorphs. This is important because β - and γ -glycine exhibit piezoelectricity, while α -glycine does not.
- **ENERGY CONVERSION:** Usage in piezoelectric nanogenerators (PENGs) and piezo-triboelectric nanogenerators (PTENGs).

SMALLEST BUILDING BLOCK, BIGGEST COSMIC STORY

For centuries, we have looked to the stars and wondered:

are we truly alone?

With the discovery of glycine—a building block of life on Earth—in comets and meteorites beyond our planet, one question remains: what else is out there?



In 2009, NASA's Stardust mission found glycine in material returned from Comet Wild 2, following earlier reports in 2003 of possible glycine in star-forming regions. Glycine has also been identified in meteorites that reached Earth. These discoveries raised a fascinating question: if glycine can exist in space, could it have survived on planets such as Mars?

In a 2014 NASA Astrobiology study, researchers tested exactly that. They trapped glycine in ice and exposed it to high-energy radiation under temperatures similar to those found on Mars. The results showed that radiation can destroy glycine, but ice can protect it. The deeper it is buried, the longer it can potentially survive—possibly for billions of years at around two meters below the Martian surface. This means that if Mars once had life, some of its chemical traces may be better preserved underground than on the surface.



REFERENCES

Aaron Gronstal Source astrobio.net. (2026). NASA Astrobiology. <https://astrobiology.nasa.gov/news/destroying-glycine-in-ice/>

Glycine. (2025). Webmd.Com. <https://www.webmd.com/vitamins-supplements/glycine#uses>

Nascimento, L., Richardson, G., et. al., (2025). The hidden power of glycine: A small amino acid with huge potential for piezoelectric and piezo-triboelectric nanogenerators. Chemical Engineering Journal, 161614. <https://doi.org/10.1016/j.cej.2025.161614>

Organics probably formed easily in early solar system. (2026). ScienceDaily. <https://www.sciencedaily.com/releases/2012/03/120330205815.htm>